
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) models have become central to scaling large language models, yet their mechanistic differences from dense networks remain poorly understood. Prior work explains dense models through *superposition*—representing more features than dimensions—and shows that superposition is governed by feature sparsity and feature importance. We show that this lens does not transfer to MoEs, and we make four contributions that establish a mechanistic account tailored to them. **First**, we identify *network sparsity* (the ratio of active to total experts), rather than feature sparsity or feature importance, as the variable that characterizes MoE representations: neither feature sparsity nor feature importance produces the discontinuous phase changes seen in dense models. **Second**, we develop new metrics that measure superposition at the level of individual experts. **Third**, using these metrics we demonstrate that models with greater network sparsity are consistently more *monosemantic*: MoEs exhibit less superposition than dense models with matched active and total parameters, with individual experts representing features more cleanly. **Fourth**, we propose a new, interpretability-focused definition of expert specialization based on monosemantic feature representation rather than load balancing, and show that experts naturally organize around coherent feature combinations when initialized appropriately. Together, these contributions indicate that network sparsity may make MoEs more interpretable without sacrificing performance, challenging the common assumption that interpretability and capability are fundamentally at odds.

1 INTRODUCTION

Mixture of Experts (MoEs) have become prevalent in state-of-the-art language models, such as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), primarily for their computational efficiency and performance gains (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work improves routing (e.g., Expert-Choice routing) and training stability/transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical results further show that learnable routers can discover latent cluster structure in data, providing insight for why experts specialize (Dikkala et al., 2023). Despite this widespread adoption, however, MoEs remain poorly understood from a mechanistic interpretability perspective—the gap this paper sets out to close.

Interpretability-oriented approaches have sought to make expert behavior more transparent, but they largely do so by changing the architecture. Yang et al. (2025b) propose MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing via expert activation. What remains missing—and what we provide—is a mechanistic account of how MoEs represent features, how experts affect superposition, and whether experts specialize without extra regularization. Mu & Lin (2025) survey MoE research and single out exactly this mechanistic understanding as a key open challenge.

The central obstacle to interpreting any neural network is superposition: the phenomenon whereby models represent more features than they have dimensions. Superposition lets

networks pack many sparse features into fewer neurons, but at the cost of making individual neurons polysemantic and hard to interpret.

MoE architectures introduce a new dimension to this problem, and it is the dimension around which our contributions are organized: network sparsity. Unlike dense models that activate all neurons regardless of input, MoEs activate a fraction of their total parameters (Shazeer et al., 2017). Where dense models exploit feature sparsity by packing many sparse features into shared neurons, MoEs can afford to be more selective, potentially dedicating entire experts to specific feature combinations—a difference we show fundamentally changes how features are represented.

We organize our study around three questions. (1) Does a MoE exhibit less superposition than its dense counterpart? (2) Does a particular feature in a MoE expert undergo a discrete phase change in superposition across its relative importance and overall feature sparsity, as seen in dense models? (3) Can expert specialization be understood through the lens of feature representation rather than load balancing alone?

We answer these questions using simple models that extend Elhage et al. (2022)’s framework to MoEs, and in doing so make three key contributions:

- **No phase changes (Section 5).** Unlike dense models, MoEs do not exhibit sharp phase changes; instead they show increasingly continuous transitions as network sparsity increases.
- **Greater monosemanticity (Section 4).** MoEs consistently exhibit greater monosemanticity—less superposition—than dense models with equivalent active and total parameters, with individual experts representing features more cleanly. We introduce per-expert metrics of superposition to establish this.
- **A feature-based definition of specialization (Section 6).** We propose an interpretability-focused definition of expert specialization based on monosemantic feature representation, and show that experts naturally organize around coherent feature combinations rather than arbitrary load balancing.

We return to each of these contributions in the sections indicated, and revisit all three together in the conclusion.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks represent concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features often exceeds the available dimensions. Elhage et al. (2022) formalized this phenomenon as superposition, demonstrating that dense models rely on non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. While superposition is theoretically efficient (Scherlis et al., 2025), it necessitates complex post-hoc disentanglement methods, such as Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). Recent work has examined how data correlations shape superposition (Prieto et al., 2025), and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). Our first two contributions extend this line of inquiry to MoE architectures: we show that network sparsity—rather than feature sparsity alone—governs representational strategies, and that this shift makes MoEs more monosemantic than matched dense models.

Interpretability of MoEs. While MoEs have become the standard for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations lags behind that of their dense counterparts. Existing analysis largely targets macroscopic behaviors, such as routing stability (Zoph et al., 2022), expert choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused approaches attempt to force specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b) or scaling expert counts to match vocabulary sizes (Park et al., 2025). These approaches,

however, prioritize capability editing over explaining intrinsic feature geometry. Our contributions directly target this gap: we analyze how experts affect superposition, show that MoEs are more monosemantic than dense models, and propose a feature-based definition of expert specialization that does not require architectural intervention.

3 BACKGROUND

A primary goal of Mechanistic Interpretability is to reverse engineer neural networks; one route is to decompose model representations into a set of human-interpretable concepts named ‘features’. These features are commonly assumed to be linear—that is, any hidden state h can be written as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ is the direction corresponding to feature i , α_i is the activation strength of this feature (roughly, the degree to which feature i is present in the input), and F is the set of all represented features.

The *superposition hypothesis* contends that models can represent far more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). To fit many more features than dimensions into a latent space, the feature vectors in F must be packed so that they are not all orthogonal. When two features have interference $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is acceptable so long as features are sparsely active (a characteristic of most online text data), but it comes at the cost of interpretability, because α_i then carries spurious activations unrelated to whether feature i is present in the input.

Monosemanticity, by contrast, is a property of individual neurons: a neuron’s activation cleanly corresponds to a single α_i (i.e., features are basis-aligned). When features are orthogonal (not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. This paper focuses on reducing superposition rather than enforcing basis-alignment—a scoping choice that matters for interpreting our contributions. For brevity, when we call a model, expert, or feature “more monosemantic,” we mean it displays less superposition.

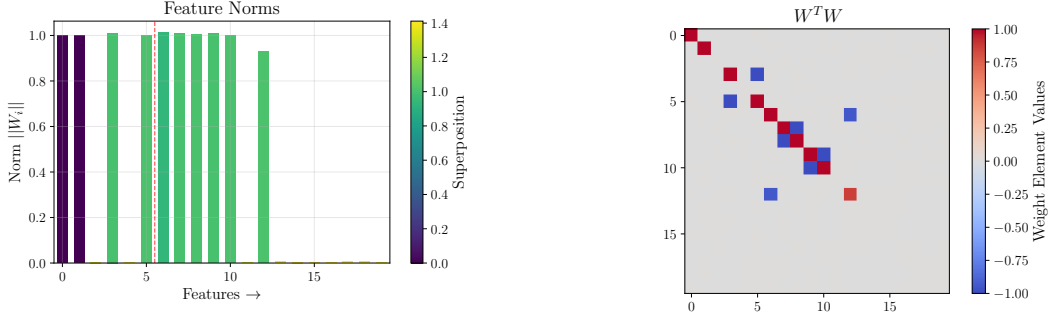
4 DEMONSTRATING SUPERPOSITION

This section develops our second contribution: that MoEs are more monosemantic than matched dense models, and the per-expert metrics needed to show it. MoEs are often conceptualized as compositions of dense models, with each expert behaving like an independent dense network. Whether experts actually represent features the way dense models do, however, remains unclear. We test this directly by comparing how MoEs and dense models differ in superposition.

4.1 EXPERIMENTAL SETUP

Our goal is to study how a MoE projects a high-dimensional vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and then accurately recovers it. This setup extends the framework of Elhage et al. (2022) to the MoE setting, and it is the common substrate for all three contributions.

The input distribution. The input vector x represents the activations of an idealized, disentangled model in which each dimension x_i corresponds to a distinct, independent feature—effectively a perfectly neuron-aligned, monosemantic representation. We take $x_i \sim U(0, 1)$, except that for a given sparsity $S \in [0, 1]$ we set $P(x_i = 0) = S$. Not all features contribute equally to the loss: to model that features vary in utility, we assign each feature x_i a scalar importance I_i that weights the reconstruction loss. To isolate and study phase transitions for a single feature, we additionally scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. Together, I_i and r vary the signal strength of the features, letting us probe expert sensitivity to feature magnitude.



(a) Norm of each feature’s weight vector $\|W_i\|$, with colors indicating superposition status (green for features in superposition, purple for monosemantic features).

(b) $W^T W$ matrix where each cell represents $(\hat{W}_i \cdot W_j)$, revealing interference patterns between features.

Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, with importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Superposition (color) is given by $\sum_j (\hat{W}_i \cdot W_j)^2$. This dense baseline is the reference against which we establish the greater monosemanticity of MoEs.

Model Architecture. The MoE consists of E experts, where each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. Inputs are routed to the top- k experts via a learned router $g(x) = \text{softmax}(W^r x)$ with $W^r \in \mathbb{R}^{E \times n}$. Each active expert projects the input to a lower-dimensional hidden state $h^e = W^e x$ and generates a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^T h^e + b^e)$ ¹. The final output is the weighted sum of the active experts: $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ where w_e are the renormalized gating weights.

We train our models with an L2 reconstruction loss weighted by the feature importances I_i , given by $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, to prevent expert collapse, we add the standard auxiliary load balancing loss (Fedus et al., 2022), defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ controls the penalty strength. Crucially, we deliberately omit auxiliary load balancing in Section 4 so that the monosemanticity we report reflects the intrinsic architectural bias of the MoE, not a regularizer we imposed.

4.2 MEASURING FEATURE CAPACITY

To make our monosemanticity claim measurable, we quantified two fundamental properties of the features: *representation strength* and *interference* with other features. We measured the norm $\|W_i^e\|$ of a feature’s weight vector in expert e , which captures the extent to which the feature is represented within that expert: $\|W_i^e\| \approx 1$ if feature i is fully represented in expert e and zero if it is not learned. We visualize the interference of feature i with other features in expert e using the Gram matrix $W^T W$, whose off-diagonal elements are the pairwise interferences. These two per-expert quantities are the metrics referenced in our second contribution.

The comparison is direct evidence for that contribution. As shown in Figures 1a and 2a, the dense model and the MoE represent a comparable number of features (10 vs. 8) with similar norms for equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$). While the MoE experts exhibit some local superposition (e.g., Expert 0 in Figure 2b), their global interference structure is strictly partitioned. Where any feature can interfere with any other in the

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^T W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

dense model (Figure 1b), the MoE enforces a block-diagonal structure in which features routed to different experts have zero interference. MoEs therefore allocate representational capacity by *partitioning* the feature space, sharply reducing the global scope of interference.

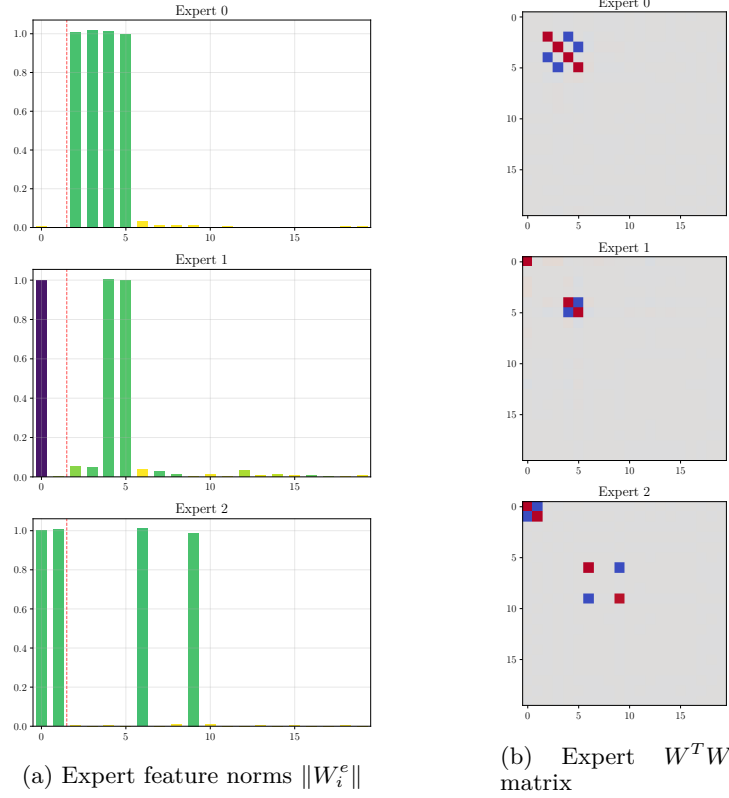


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (top- $k = 1$ routing), with importance $I = 0.7^i$ and feature density $1 - S = 0.1$. Compared with the dense baseline of Figure 1, interference is confined to a block-diagonal structure—the visual signature of the greater monosemanticity we claim.

Expert Feature Dimensionality. To understand how MoEs allocate their limited representational capacity differently from the dense model, we measured feature dimensionality: the “fraction of a dimension” that a specific feature receives in a model (Elhage et al., 2022). This is the second per-expert metric underpinning our monosemanticity contribution. For a feature i , we define its dimensionality in expert e by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero (not learned) and one (monosemantic). The total capacity of a MoE can then be written as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When features are “efficiently packed” in a model’s representation space, their dimensionalities sum to the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE the relation becomes $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this constraint, so MoEs pack features just as efficiently as dense models at the same total parameter count. This rules out the alternative explanation that MoEs are simply less capable of using their capacity, and lets us attribute the difference we report to *how*, not *how much*, capacity is used.

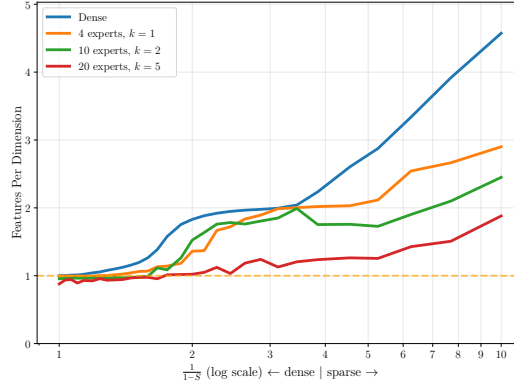


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-g}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). This figure is the headline evidence for our monosemanticity contribution: the dense model ($n = 100$, $m = 20$) shows the most superposition, and superposition falls monotonically as the expert count rises—4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameters and similar k/E . The dashed line at 1.0 marks fully monosemantic representation.

Features per Dimension & Network Sparsity. Because both dense and MoE models “efficiently pack” features, we can compare them directly on a single axis: features per dimension. This quantity lets us measure superposition exactly in both architectures and trace how the number of experts in a MoE affects superposition across feature sparsities. When features per dimension exceeds one, the model is in superposition—it represents more features than it has dimensions. We define features per dimension for a MoE by

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected probability that expert e is used across a batch of input samples, i.e. the average renormalized gating weight after top- k routing.

The results confirm our second contribution. For MoEs with an *equal* number of total parameters as the dense model, the dense model has a higher number of features per dimension (Figure 3)—that is, more superposition. The dense model uses superposition to represent a greater total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE caps its representation near the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Moreover, as we increase the total number of experts—holding the total parameters and the ratio k/E roughly fixed—features per dimension falls further. In short: *the greater the number of experts, the less superposition in the model*, and correspondingly the more monosemantic its features. This monosemanticity is not free of a performance cost, but the cost is small and shrinks with scale: the dense model’s extra superposition buys it consistently lower reconstruction loss than the MoEs (Figure 6, Appendix A.1), with a gap of only ~ 0.03 – 0.08 at any given sparsity, and as the number of experts grows the MoEs close that gap. Appendix A.2 gives the theoretical intuition for why network sparsity drives this reduction in superposition.

5 PHASE CHANGE

This section establishes our first contribution: unlike dense models, MoEs do not undergo discrete phase changes. The previous section showed that although MoEs and dense models learn a similar number of features, MoEs distribute them across experts with less interference, suggesting that network sparsity reshapes *how* features are allocated rather than *how*

many are learned. Here we ask whether the input-distribution properties known to govern dense-model representation—feature sparsity and importance—induce the same physics-inspired *phase changes* in MoEs.

A model has only finitely many ways to represent a feature: it may ignore it, superimpose it, or represent it monosemantically. A phase change is the observation that discrete boundaries sometimes separate these regimes as functions of feature sparsity and relative importance.

Dense toy models exhibit exactly such discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022): varying the sparsity and relative importance of input features elicits different behavior, with greater feature sparsity encouraging greater superposition. Our contribution is to show that the per-expert phase diagram of a MoE looks qualitatively different, revealing a distinct representational strategy.

Setup. We follow the setup of Section 4.1, adding a load balancing loss to avoid specialization collapse—the failure mode in which experts and routers fall into local minima where certain experts are entirely ignored—in setups where it occurs. We study three configurations, all with one active expert ($k = 1$): (A) $n = 2, m = 1$; (B) $n = 3, m = 1$; and (C) $n = 3, m = 2$. For each, we report the expert-specific phase diagram across feature sparsity and last-feature relative importance, and we sweep network sparsity by increasing the number of experts (E) up to the number of input feature dimensions (n).

Here we fix active parameters rather than total parameters, so that $m_{dense} = km_{moe}$ (ignoring router parameters). This choice lets us compare within a model architecture; otherwise any observed difference could be attributed to architectural changes rather than to the number of active parameters. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2, but the latter has only one hidden dimension ($m = 1$) to encode the same number of input features ($n = 3$) that the former encodes with two ($m = 2$), making it hard to use superposition to reason about specialization.

Results and Takeaways. The results deliver our first contribution directly. In every single-expert (dense) case we observed a clear phase change (Figure 4.X.1/1), affirming Elhage et al. (2022). As soon as we increased the total number of experts, discrete phase changes disappeared. Some experts in MoEs with $E = 2$ resemble their respective dense cases (Figure 4.X.2/2) but transition more continuously, and in each case one expert became more monosemantic, specializing in the most important feature by relative importance. Experts unlike the dense cases uniformly have much lower superposition scores (they are bluer), i.e. more monosemantic representations. This dovetails with the previous section’s finding: MoEs favor lower superposition than their dense counterparts, whether we measure it globally (Section 4) or through the phase diagram here.

The individual configurations reinforce the same point. For the $n = 2, m = 1$ setup (Figure 4.A), the dense model drops the last feature when feature sparsity is low, whereas the comparable MoE preserves it far more often because it has the capacity to do so. With three input dimensions (Figure 4.B) the MoEs no longer show this behavior, because the experts are already superimposing the other two features and there is no room for the third within a single hidden dimension; here the superimposed hidden dimension is, in the high-sparsity regime, insufficient to represent all features. Even so, we see no clear phase change—except in the 0.5–0.7 feature-sparsity region of 4.B.2/2, which is mostly discrete but mixed. For $m = 2$, the white region of the dense model (Figure 4.C.1/1), in the mid- to low-sparsity domain where the last feature is relatively less important, ignores that feature. As network sparsity increases—across all other panels of Figure 4.C—the models represent the last feature with greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Equivalently, dimensionality in the low relative-importance region rises with network sparsity, consistent with Figure 3.

We also observed a window of feature sparsity from roughly 0.48 to 0.7 in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 with a heavy mix of polysemanticity, monosemanticity, and outright ignorance. This indicates a middle ground in which polysemantic and monosemantic representations achieve comparable loss, making it hard for the MoE to commit consistently to the strategies seen at low and high feature sparsity. Dense models show no such window—further evidence, in support of our first contribution, that MoEs learn qualitatively different representational strategies.

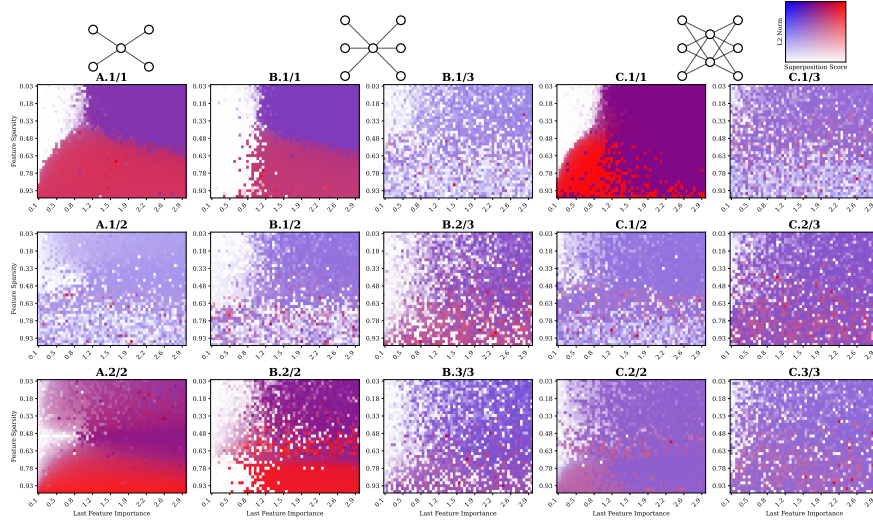


Figure 4: Phase diagrams supporting our first contribution—the disappearance of discrete phase changes in MoEs. For a particular expert and input dimension (feature), we decode how the feature is embedded in the hidden dimension: ignored (white), monosemantic (blue-purple), or superimposed (red). We plot the joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j<n}(\hat{W}_n \cdot W_j)^2$) across feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last feature of n total features. For each cell we train ten models and select the one with the lowest loss, and we use the load balancing loss throughout this section. A low L2 norm ($\|W_n\|$) is white, denoting that the model ignores the last feature; otherwise a low superposition score is blue-purple, indicating monosemantic representation, and red indicates the feature is represented in superposition. Cell (i, j) in subfigure X.e/E denotes expert e of E total experts trained on architecture X for last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 is the dense model. The sharp boundaries in the X.1/1 (dense) column give way to smoother transitions as E grows.

Takeaway. All of these experiments use top- $k = 1$, so only one expert is ever active at a time. Even under this most dense-like setting, behavior changes dramatically as early as $E = 2$ —including when superposition within the hidden dimension would suffice to represent all features. We therefore conclude, in support of our first contribution, that it is misleading to view a MoE as an aggregation of dense models: the router, by letting each expert see only a subset of the feature domain, fundamentally alters how experts behave and what they learn.

6 EXPERT SPECIALIZATION

This section develops our third contribution: a feature-based definition of expert specialization. Having established that MoEs exhibit less superposition, we now examine how the resulting monosemantic features are organized within experts and how that organization relates to specialization.

Expert specialization in MoEs is traditionally framed around load balancing between experts across all inputs (Chaudhari et al., 2025). That framing, however, misses the natural intuition of specialization—that an expert should be used precisely when the concepts it is specialized in are present in the input.

We therefore propose a different definition: an expert is *specialized* if it *occupies* certain feature directions in the input space *and* represents those features relatively monosemantically. Our contribution is to show that these two conditions are directly correlated, and that their joint presence itself encourages load balancing across experts—recovering the classical property as a consequence rather than an objective.

We first make the notion of *occupancy* precise. Because we fix $k = 1$, the feature space is partitioned into convex cone regions (see Appendix A.3), each routed to a particular expert. By definition $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points in the cone and s is any positive scalar. Thus if a feature vector $\vec{x}$ is routed to an expert, so is every $s\vec{x}$. In this case we say the feature x is contained within expert e , and that expert e *occupies* x .

Empirically, small models that distribute the input space across more experts tend to achieve lower loss (Appendix A.4). This raises the question that motivates our definition: does allocating the input space to certain experts imply anything about those experts? Our definition predicts that it implies *monosemanticity*—a correlation we confirm below and show holds even for larger toy models (e.g., $m = 10$).

To test the two directions of this correlation, in models with $n > 2$ we ask (i) whether initializing experts to occupy features in the input space causes those experts to be more monosemantic with respect to those features, and (ii), conversely, whether the features an expert chooses to represent monosemantically are the ones it occupies in the input space.

The initialization sweep confirms direction (i). When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and only that feature (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the other experts, initialized over less important features, represented neither those features nor the five most important features they were not initialized over. When we break the importance ordering and randomize the features each expert initially occupies, each expert monosemantically represented only the most important feature(s) it was initialized over (Figure 5c).

Two regularities emerge, both central to our definition. First, there is a strong correlation between the features initially routed to an expert and the features that expert comes to represent monosemantically. Second, experts monosemantically represent *only* important features—whether we hand an expert one important feature explicitly or a set of features from which it selects the most important itself.

Even under uniform feature importance, where experts place all features in superposition with higher interference, the features an expert initially occupies remain *relatively* more monosemantic, achieving superposition scores on average 1.03 standard deviations below that expert’s mean. Occupancy thus predicts monosemanticity regardless of the importance profile.

We next tested direction (ii): whether experts that represent certain features monosemantically also occupy those features in the input. We measure usage statistics when those features are *one of many* active features and when they are the *only* active features—the latter being equivalent to the probability that the expert occupies the features. The correlation holds under both xavier and k-hot initialization (Table 1). Since $E = 10$, a mean expert usage of $\sim 10\%$ indicates even load balancing across experts. In every case, activating an expert’s monosemantic feature(s) raises its usage substantially, and making that feature the only active one lets the expert dominate. Most strikingly, under k-hot initialization *100%* of all features an expert represents monosemantically are occupied by that same expert—the two conditions of our definition coincide exactly.

Taken together, these results support our third contribution: as an expert represents more features monosemantically, it becomes more specialized in exactly our sense. Its usage on arbitrary input falls, but conditional on its specialized features being active its usage rises far above that of other experts. This pattern holds in every case except the xavier-initialized expert with four monosemantic features, where utilization drops sharply—the lone exception to an otherwise consistent trend.

7 CONCLUSION

This paper set out to give MoEs a mechanistic account of their own, and it delivered three contributions toward that goal. First, we showed that MoEs do not undergo the discrete

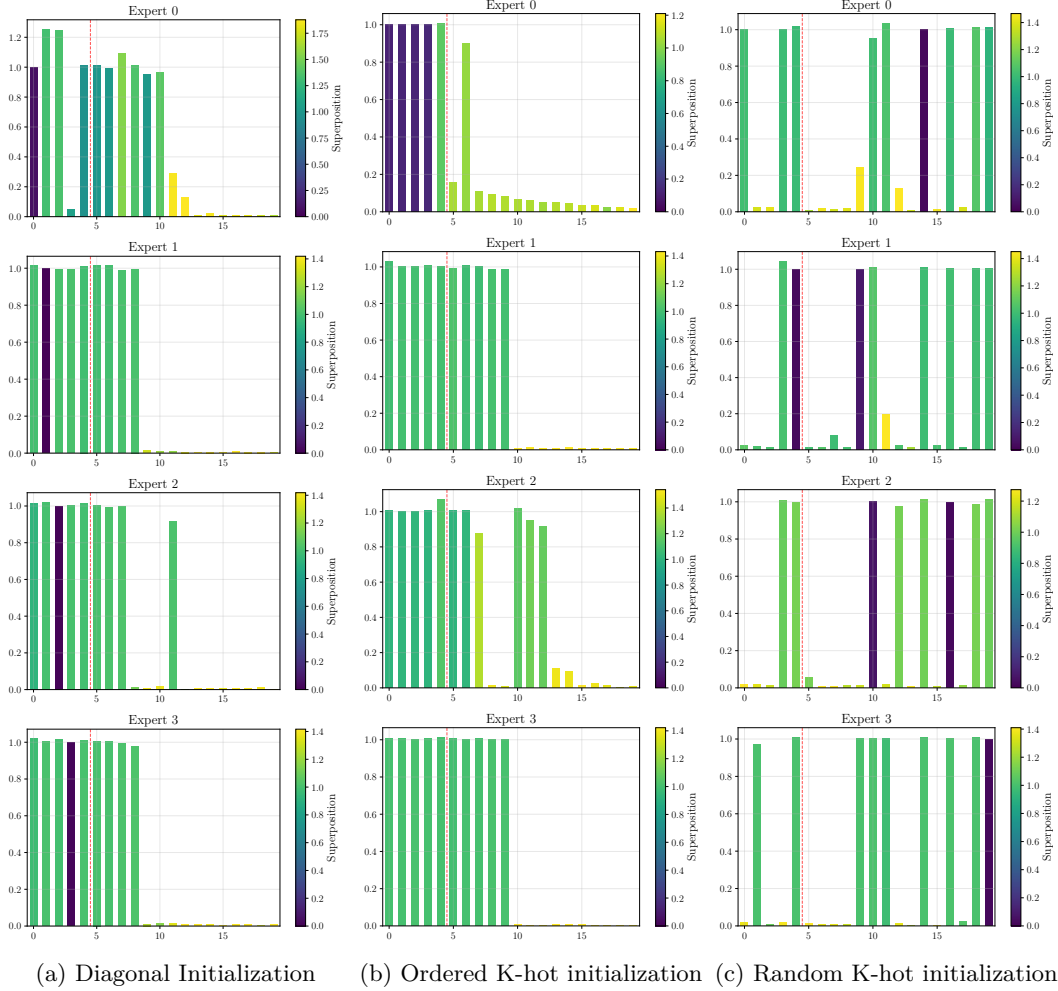


Figure 5: Evidence for our specialization contribution: expert feature norms $\|W_i^{(e)}\|$ and superposition (color) under three initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. Across all three, the features an expert initially occupies are the ones it comes to represent monosemantically. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially from feature one to 20. In (b), the gate matrix is initialized “ordered k-hot,” so the first expert aligns with the first five features and each subsequent expert with the next five; relative feature importance is as in (a). In (c), the gate matrix is initialized “random k-hot,” with each expert assigned five random features such that experts share no common feature yet collectively cover all 20; relative feature importance still decreases exponentially but is randomly distributed across features.

phase changes seen in dense models, transitioning instead ever more continuously as network sparsity grows. Second, using new per-expert metrics of superposition, we showed that MoEs are consistently more monosemantic than dense networks with matched parameters. Third, we proposed a feature-based definition of expert specialization and demonstrated that, when initialization encourages it, experts naturally organize around coherent feature combinations—recovering load balancing as a byproduct rather than an objective. Our findings rest on simple autoencoder toy models with synthetic data, and so leave open how they generalize to large-scale transformers with unknown feature distributions (Appendix A.6). Even within these limits, the through-line across all three contributions is that toy MoEs achieve comparable loss while remaining more interpretable—challenging the prevailing view that mechanistic interpretability and model capability are fundamentally in tension. Build-

Table 1: Monosemantic-feature and usage statistics per expert for $n = 100, m = 10, E = 10$, quantifying the occupancy–monosemanticity link at the heart of our specialization contribution. We train one hundred models for each initialization scheme (xavier and k-hot), giving 1000 experts each. Statistics are aggregated across models, with experts grouped by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we report its usage when those feature(s) are one of several active features in the input, and its usage when they are the *only* active feature. Usage rises sharply with both conditions, and reaches 100% once the feature is the sole active input.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	–	–	–
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	–	–	–
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	–	–	–
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

ing on these contributions, future work should ask what favors monosemanticity in MoEs, how their training dynamics differ from dense models’, and when specialization emerges—questions whose answers could inform the design of more interpretable, high-performing language models.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

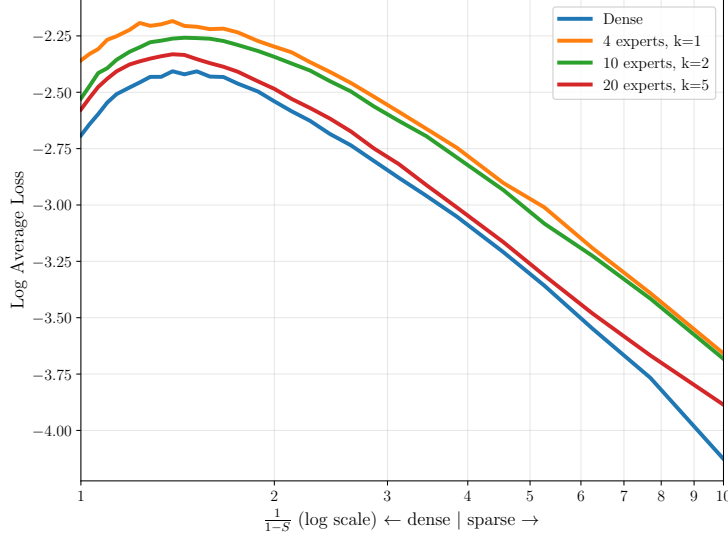


Figure 6: Log average loss versus feature density ($\frac{1}{1-S}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. This figure quantifies the modest performance cost behind our monosemanticity contribution: the dense model outperforms every MoE at each sparsity level, but the gap narrows steadily as the number of experts grows.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

This appendix explains *why* our monosemanticity contribution holds. In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs “eat this same gap” structurally, through conditional computation: by aligning activation sparsity with feature sparsity, they remove the computational penalty for maintaining dedicated, rarely-activating neurons. As Elhage et al. (2022) note, once a model only spends computation on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes optimal—trading the *compression* of superposition for the *selection* of routing.

Geometrically, this structural change shows up as reduced *interference*. Although the global ratio of features represented to parameters stays constant, the router partitions the feature space so that a feature routed to expert e competes for capacity only with the subset of features also assigned to e . Interference is therefore governed by the local expert matrix $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, easing the optimization pressure to pack features in superposition and letting experts learn monosemantic representations—precisely the effect we measure in Section 4.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

This appendix justifies the occupancy geometry underlying our specialization contribution. In the regime $k = 1$, the router $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$. The region routed to expert i is $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are row vectors of W^r —a homogeneous linear inequality. Regions bounded by such inequalities are, by

definition, convex cones: if a given x satisfies the inequality, so does sx for any positive scalar s , and if x_1 and x_2 both satisfy it, so does every $x = x_1\lambda + (1 - \lambda)x_2$ with $\lambda \in [0, 1]$.

For $k > 1$, the region of inputs routed to a particular expert e becomes a union of convex cones, which is generally not itself a convex cone. Our notion of experts occupying feature directions therefore may not extend beyond $k = 1$ —a scope condition we flag explicitly so the specialization contribution is not overread.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

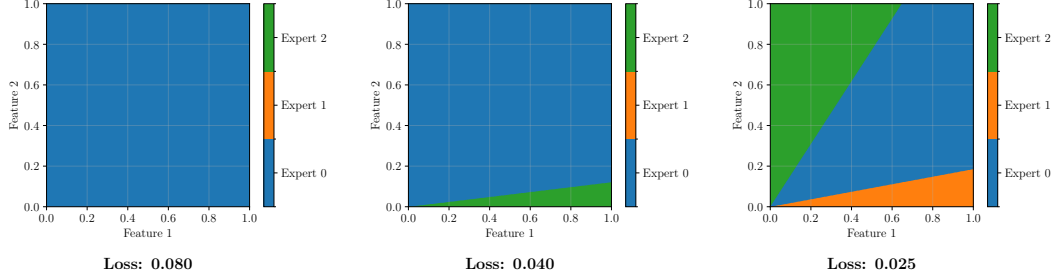


Figure 7: Expert routing of three identical models under different initialization schemes ($n = 2$, $E = 3$, $m = 1$), illustrating the link between input-space distribution and performance that motivates our specialization contribution. The first model (left) performs worst (loss: 0.08) and routes all inputs to one expert. The second (middle) does better (loss: 0.04), routing a small slice of inputs—those with feature 1 active—to a second expert. The third (right) achieves the lowest loss (0.025) and distributes the input space across all experts: one expert fires when only feature 1 is active, one when only feature 2 is active, and one when both are.

In small models ($n = 2, m = 1, E > 1$), we find empirically that distributing the input space across more experts tends to lower loss, tested over $E \in [2, 7]$. Holding $n = 2$ lets us visualize which portions of the input space route to which experts (Figure 7).

A.5 ANALYTIC MODEL EQUIVARIANCE

This appendix supplies a theoretical complement to our monosemanticity contribution: for the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a map from any dense model to a monosemantic MoE with an equivalent number of active features under a sparsity constraint.

Assume an upper bound a on the number of active features for any input, so that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Assume further that a is no greater than an expert’s hidden dimensionality m —bounding how many features a model must represent—and that the hidden dimension is smaller than the total number of input features n (i.e., $a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —meaning the size of the subset of active features is smaller than or equal to a —create an expert which monosemantically preserves those features. (In fact, you can take only the subsets such that $|S| = a$.) The router then selects the expert which corresponds to those active features (of which there will never be more than a , by assumption):

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has enough capacity to represent its assigned features without superposition. In other words, only a features are ever active, and every unique combination of active features gets its own dedicated expert with capacity to represent those features monosemantically. The number of experts this construction may require is thus $\binom{n}{a}$.

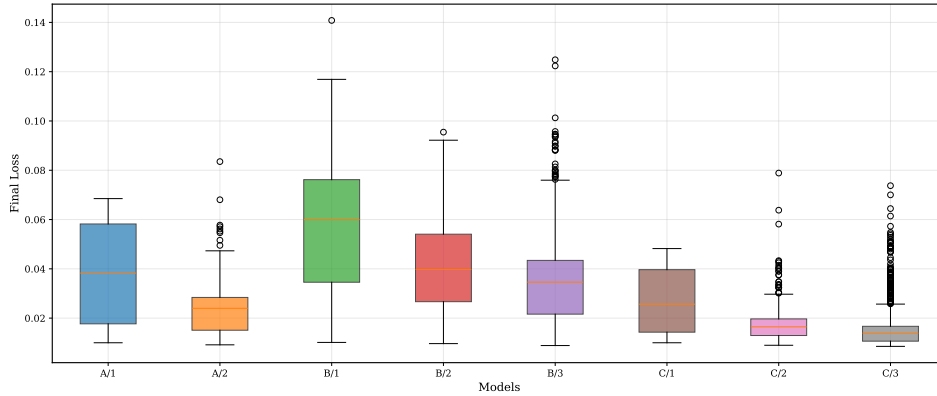


Figure 8: Model X/E reuses the architectures of Figure 4, with E the total number of experts (i.e., network sparsity). Increasing network sparsity lowers mean loss while raising localized variance—most sharply as the expert count approaches the input feature dimensions. We attribute the added variance to the relatively unstable training of MoEs compared to dense models, which persists even though we train ten models per cell and keep the lowest loss.

This theoretical MoE reconstructs with zero loss only while the sparsity constraint holds (i.e., as sparsity goes to one in these toy models), because more than m features could occasionally be active at once ($a \not\leq m$), exceeding the network’s monosemantic capacity. Crucially, though, a dense polysemantic model faces the very same problem unless features are correlated in the distribution—so even when $a \not\leq m$ sometimes, the monosemantic MoE under this construction may still outperform the dense model under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE $f_{\text{MoE}}(x)$ with $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs; in this paper’s toy settings, that constraint holds in the limit where sparsity goes to one. In practice, moreover, there may be an upper bound on how many features a given amount of information can semantically encode, reflected in the size of meaningful embeddings of that data—so a MoE with enough experts and a tractable, interpretable amount of superposition may suffice to encode all features present.

A.6 LIMITATIONS

Our three contributions should be read in light of several limitations that bound their scope. First, all experiments live in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This enables precise measurement of interference and monosemanticity—the basis of our metrics contribution—but abstracts away much of what large-scale transformers involve, including multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics. How well our findings transfer to real-world MoEs therefore remains uncertain.

Second, our architectural and routing choices are intentionally restricted: simple experts, fixed top- k routing (often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs often use richer routing, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or move beyond reconstruction to next-token prediction—any of which could yield qualitatively different representational behavior, and in particular could alter the phase-change and specialization claims.

Third, the feature distributions in our synthetic tasks—their sparsity patterns, feature importance, and independence assumptions—are far simpler than those in natural data.

Finally, although we observe increased monosemanticity and reduced superposition under fixed conditions, we do not evaluate downstream performance trade-offs, training stability,

or specialization dynamics at scale; prior work suggests these can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

In sum, our study establishes its contributions under clean experimental conditions, but validating them in large, realistic MoE systems requires further work. Our methods do not scale naturally to larger models, which we see as the clearest direction for future work.